# CSGE602055 Operating Systems
# CSF2600505 Sistem Operasi
# Week 05: Virtual Memory

C. BinKadal

Sendirian Berhad

https://os.vlsm.org/Slides/os05.pdf
Always check for the latest revision!

REV386 28-Jul-2022

# OS222[3]): Operating Systems Schedule 2022 - 2

| Week | Topic[1]) | OSC10[2]) |
|---|---|---|
| Week 00 | Overview (1), Assignment of Week 00 | Ch. 1, 2 |
| Week 01 | Overview (2), Virtualization & Scripting | Ch. 1, 2, 18. |
| Week 02 | Security, Protection, Privacy, & C-language. | Ch. 16, 17. |
| Week 03 | File System & FUSE | Ch. 13, 14, 15. |
| Week 04 | Addressing, Shared Lib, & Pointer | Ch. 9. |
| Week 05 | Virtual Memory | Ch. 10. |
| Week 06 | Concurrency: Processes & Threads | Ch. 3, 4. |
| Week 07 | Synchronization & Deadlock | Ch. 6, 7, 8. |
| Week 08 | Scheduling + W06/W07 | Ch. 5. |
| Week 09 | Storage, Firmware, Bootloader, & Systemd | Ch. 11. |
| Week 10 | I/O & Programming | Ch. 12. |

[1]) For schedule, see https://os.vlsm.org/#idx02

[2]) Silberschatz et. al.: **Operating System Concepts**, $10^{th}$ Edition, 2018.

[3]) This information will be on **EVERY** page two (2) of this course material.

# STARTING POINT — https://os.vlsm.org/

- **Text Book** — Any recent/decent OS book. Eg. (**OSC10**) Silberschatz et. al.: **Operating System Concepts**, $10^{th}$ Edition, 2018. (See https://www.os-book.com/OS10/).
- **Resources** (https://os.vlsm.org/#idx03)
  - **SCELE OS222** — https://scele.cs.ui.ac.id/course/view.php?id=3398. The enrollment key is **XXX**.
  - **Download Slides and Demos from GitHub.com** — (https://github.com/os2xx/os/) os00.pdf (W00), os01.pdf (W01), os02.pdf (W02), os03.pdf (W03), os04.pdf (W04), os05.pdf (W05), os06.pdf (W06), os07.pdf (W07), os08.pdf (W08), os09.pdf (W09), os10.pdf (W10).
  - **Problems** 195.pdf (W00), 196.pdf (W01), 197.pdf (W02), 198.pdf (W03), 199.pdf (W04), 200.pdf (W05), 201.pdf (W06), 202.pdf (W07), 203.pdf (W08), 204.pdf (W09), 205.pdf (W10).
  - **LFS** — http://www.linuxfromscratch.org/lfs/view/stable/
  - **OSP4DISS** — https://osp4diss.vlsm.org/
  - **This is How Me Do It!** — https://doit.vlsm.org/001.html
    - PS: "Me" rhymes better than "I", duh!

# Agenda


1. Start
2. OS222 Schedule
3. Agenda
4. Week 05
5. OSC10 (Silberschatz) Chapter 10: Virtual Memory
6. Virtual Memory
7. Memory Allocation Algorithm
8. TOP: Table of Processes

# Week 05 Virtual Memory: Topics[1]

- Review of physical memory and memory management hardware
- Virtual Memory
- Caching
- Memory Allocation
- Memory Performance
- Working sets and thrashing

[1] Source: ACM IEEE CS Curricula 2013

# Week 05 Virtual Memory: Learning Outcomes[1]

- Explain memory hierarchy and cost-performance trade-offs. [Familiarity]
- Summarize the principles of virtual memory as applied to caching and paging. [Familiarity]
- Describe the reason for and use of cache memory (performance and proximity, different dimension of how caches complicate isolation and VM abstraction). [Familiarity]
- Defend the different ways of allocating memory to tasks, citing the relative merits of each. [Assessment]
- Evaluate the trade-offs in terms of memory size (main memory, cache memory, auxiliary memory) and processor speed. [Assessment]
- Discuss the concept of thrashing, both in terms of the reasons it occurs and the techniques used to recognize and manage the problem. [Familiarity]

[1] Source: ACM IEEE CS Curricula 2013

# OSC10 (Silberschatz) Chapter 10: Virtual Memory

- OSC10 Chapter 10
  - Background
  - Demand Paging
  - Copy-on-Write
  - Page Replacement
  - Allocation of Frames
  - Thrashing
  - Memory-Mapped Files
  - Allocating Kernel Memory
  - Other Considerations
  - Operating-System Examples

# Virtual Memory

- Virtual Memory: Separation Logical from Physical.
- Virtual Address Space: logical view.
- Demand Paging
- Page Flags: Valid / Invalid
- Page Fault
- Demand Paging Performance
- Copy On Write (COW)
- Page Replacement Algorithm
  - Reference String
  - First-In-First-Out (FIFO)
  - Belady Anomaly
  - Optimal Algorithm
  - Least Recently Used (LRU)
  - LRU Implementation
  - Lease Frequently Used (LFU)
  - Most Frequently Used (MFU)

# Memory Allocation Algorithm

- Page-Buffering Algorithms
- Allocation of Frames
- Fixed Allocation
- Priority Allocation
- Global vs. Local Allocation
- Non-Uniform Memory Access (NUMA)
- Thrashing
- Working-Set Model
- Shared Memory via Memory-Mapped I/O
- Kernel
  - Buddy System Allocator
  - Slab Allocator

# TOP: Table of Processes (12-memory.c) (01)

See also https://osp4diss.vlsm.org/osp-101.html

```
/*
 * Copyright (C) 2016-2021 Rahmat M. Samik-Ibrahim
 * http://rahmatm.samik-ibrahim.vlsm.org/
 * This program is free script/software. This program is distributed in the
 * hope that it will be useful, but WITHOUT ANY WARRANTY; without even the
 * implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# INFO: TOP (Table of Processes)
 * REV11 Tue 30 Mar 18:25:50 WIB 2021
 * REV07 Fri 26 Mar 22:52:06 WIB 2021
 * REV04 Mon 12 Mar 17:33:30 WIB 2018
 * START Mon 03 Oct 09:26:51 WIB 2016
 */

#define TOKEN  "OS212W05"
#define MSTARTS 0x125E4
// #define MSTARTS 0x2BE5
// #define MSTARTS 0xFE4
// #define MSTARTS 0x3E4
// #define MSTARTS 0x1E4

#define MSIZE14 0x40609
#define MSIZE13 0x40609
#define MSIZE12 0x40608
#define MSIZE11 0x40608
#define MSIZE10 0x20FE8
#define MSIZE09 0x20FE8
#define MSIZE08 0x1F609
```

# TOP: Table of Processes (12-memory.c) (02)

```
#define MSIZE07 0x1F609
#define MSIZE06 0x1F608
#define MSIZE05 0x1F608
#define MSIZE04 0x1E609
#define MSIZE03 0x1E609
#define MSIZE02 0x1E609
#define MSIZE01 0x1E608
#define MSIZE00 0x1E608
#define LINE   75
#define MAXSTR 80
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/types.h>

typedef  unsigned char* uChrPtr;
void     chktoken (uChrPtr token);

void printLine(int line) {
   while(line-- > 0) putchar('x');
   putchar('\n');
   fflush(NULL);
}

uChrPtr GlobalChar[MSTARTS];
```

# TOP: Table of Processes (12-memory.c) (03)

```
void main(void) {
   int   msize[] = {MSIZE00, MSIZE01, MSIZE02, MSIZE03, MSIZE04,
                    MSIZE05, MSIZE06, MSIZE07, MSIZE08, MSIZE09,
                    MSIZE10, MSIZE11, MSIZE12, MSIZE13, MSIZE14};
   int   ii, jj;
   int   myPID    = (int) getpid();
   char  strSYS1[MAXSTR], strOUT[MAXSTR];
   char* chrPTR;
   char* chrStr;

   printLine(LINE);
   printf("ZCZC chktoken\n");
   chktoken(TOKEN);
   printLine(LINE);

   sprintf(strSYS1, "top -b -n 1 -p%d | tail -5", myPID);
   system (strSYS1);
   sprintf(strSYS1, "top -b -n 1 -p%d | tail -1", myPID);
   printf("PART 1\n");
   printLine(LINE);
   for (ii=0; ii < (sizeof(msize)/sizeof(int)); ii++){
      chrStr = malloc(msize[ii]);
      FILE* filePtr=popen(strSYS1, "r");
      fgets(strOUT, sizeof(strOUT)-1, filePtr);
      pclose(filePtr);
      strOUT[(int) strlen(strOUT)-1]='\0';
      printf("%s [%X]\n", strOUT, msize[ii]);
      free(chrStr);
   }
```

# TOP: Table of Processes (12-memory.c) (04)

```
   printf("\nPART 2\n");
   printLine(LINE);
   for (ii=0; ii < (sizeof(msize)/sizeof(int)); ii++){
      chrPTR = chrStr = malloc(msize[ii]);
      for (jj=0;jj<msize[ii];jj++)
         *chrPTR++='x';
      FILE* filePtr=popen(strSYS1, "r");
      fgets(strOUT, sizeof(strOUT)-1, filePtr);
      pclose(filePtr);
      strOUT[(int) strlen(strOUT)-1]='\0';
      printf("%s [%X]\n", strOUT, msize[ii]);
      free(chrStr);
   }
}
```

# TOP: Table of Processes (13-chktoken.c) (05)

```
/*
 * Copyright (C) 2021 Rahmat M. Samik-Ibrahim
 * http://rahmatm.samik-ibrahim.vlsm.org/
 * This program is free script/software. This program is distributed in the
 * hope that it will be useful, but WITHOUT ANY WARRANTY; without even the
 * implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
 * REV05: Tue 30 Mar 14:55:36 WIB 2021
 * REV04: Tue 30 Mar 10:35:13 WIB 2021
 * REV03: Tue 30 Mar 08:36:56 WIB 2021
 * START: Mon 22 Mar 2021 16:14:36 WIB
 *
# INFO: chktoken(TOKEN) function
 */

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>

#define  MAXINPUT   256
#define  MAXCMD     MAXINPUT
#define  MAXOUTPUT  MAXINPUT
#define  RESULT     4

typedef           char  Chr;
typedef           char* ChrPtr;
typedef  unsigned char  uChr;
typedef  unsigned char* uChrPtr;
```

# TOP: Table of Processes (13-chktoken.c) (05)

```
#define CMDSTRING "echo %s | sha1sum | cut -c1-4 | tr '[:lower:]' '[:upper:]' "
void mySHA1(uChrPtr input, uChrPtr output) {
    Chr  cmd[MAXCMD];
    sprintf(cmd, CMDSTRING, input);
    FILE* filePtr = popen(cmd, "r");
    fgets(output, RESULT+1, filePtr);
    output[RESULT]=0;
    pclose(filePtr);
}
void getTimeStamp(uChrPtr timeStamp) {
    time_t tt    =  time(NULL);
    struct tm tm = *localtime(&tt);
    sprintf(timeStamp, "%2.2d%2.2d", tm.tm_min, tm.tm_sec);
}
void     chktoken (uChrPtr token) {
    uChr    input [MAXINPUT];
    uChr    output[MAXOUTPUT];
    uChr    timeStamp[] = "MMSS";
    uChrPtr user        = getenv("USER");
    getTimeStamp(timeStamp);
    int     len   = strlen(timeStamp);
    strcpy(input,timeStamp);
    strcpy(input+len,user);
    len          += strlen(user);
    strcpy(input+len,token);
    len          += strlen(token);
    mySHA1(input,   output);
    printf("%s %s-%s\n", user, timeStamp, output);
}
```

# TOP: Table of Processes (13-chktoken) (06)

```
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
ZCZC chktoken
cbkadal 5257-80A5
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
MiB Mem :    986.5 total,    157.1 free,    174.2 used,    655.2 buff/cache
MiB Swap:    488.0 total,    488.0 free,      0.0 used.    632.0 avail Mem

  PID    VIRT    RES    SHR   SWAP   CODE    DATA   USED nDRT
  864    6000   1528   1240      0      8     948   1528    0
PART 1
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
  864    6000   1528   1240      0      8     948   1528    0 [1E608]
  864    6000   2620   2292      0      8     948   2620    0 [1E608]
  864    6132   2620   2292      0      8    1080   2620    0 [1E609]
  864    6004   2620   2292      0      8     952   2620    0 [1E609]
  864    6004   2620   2292      0      8     952   2620    0 [1E609]
  864    6004   2620   2292      0      8     952   2620    0 [1F608]
  864    6004   2620   2292      0      8     952   2620    0 [1F608]
  864    6136   2620   2292      0      8    1084   2620    0 [1F609]
  864    6136   2624   2292      0      8    1084   2624    0 [1F609]
  864    6136   2624   2292      0      8    1084   2624    0 [20FE8]
  864    6136   2624   2292      0      8    1084   2624    0 [20FE8]
  864    6136   2624   2292      0      8    1084   2624    0 [40608]
  864    6136   2624   2292      0      8    1084   2624    0 [40608]
  864    6268   2624   2292      0      8    1216   2624    0 [40609]
  864    6264   2624   2292      0      8    1212   2624    0 [40609]
```

# TOP: Table of Processes (13-chktoken) (07)

```
PART 2
xxxx
```